

Stopped-current FAR and complete-current NEAR coordinates

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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1 Purpose and scope

This note advances the R-081 frontier without asserting the still missing Sector-A estimate. It has four positive conclusions.

First, the root-resolved FAR loss is rewritten exactly as one deterministic stopped-current square. The formula keeps the moving endpoint and both predictable control drifts; it is not an optional-stopping argument. Second, the uncontrolled production FAR contribution is closed uniformly in the cutoff by extracting a support-refined positive-regularity corollary from the R-050 paracontrolled reconstruction. Third, an exact orthogonal causal Carleson lemma identifies the strict half-derivative threshold for one factorised controlled route. Fourth, the production Q_{Π} current is put in a global four-coordinate Pauli–Fierz square, and the exact conditional-covariance criterion for a positive square–trace branch is isolated.

The note also records the load-bearing negative boundaries. The predictable drift cannot be omitted. The raw value innovation is not centered without its heat compensator. The four-coordinate Pauli–Fierz map preserves pointwise energy and the Wick Gram matrix but not the individual martingale root blocks of the fixed six-current. Conditional covariance preservation is not automatic under future feedback. Moving sharp-shell edge flux is a signed repartition, not a new positive payment. Consequently controlled FAR, the complete signed NEAR packet, overlap-stable progression, controlled-shell one-use, the $q = 10/9$ Nelson estimate, and Sector-A closure remain open.

All statements are at finite cutoff, on the pinned production model, at fixed positive density floor, with a common even regulator and exact non-aliased products. Constants declared uniform in the cutoff may depend on the floor, the fixed low root, and the indicated regularity exponents.

2 Authority and notation

The load-bearing authorities are R-050 for the enhanced Gaussian current and its first parilinearisation remainder, R-063 for the complete balanced square–trace–forest packet, R-069–R-070 for exact current telescopes and trace bookkeeping, R-075 for the production Q_{Π} frame, R-079 for the complete heat-projected current identity, R-080 for FAR square completion, and R-081 for the polynomial/Cartan split and the secant–Jensen boundary.

Let $(\mathcal{F}_j)_{j=\ell}^J$ be the root filtration, where $\ell = j_0 - 1$, and let

$$P_j = \mathbb{E}[\cdot \mid \mathcal{F}_j], \quad d_j = P_j - P_{j-1}.$$

The spatial Littlewood–Paley projector is denoted by Π_m ; it is deterministic and therefore commutes with conditional expectation. The complete current already contains the fixed factor $Q_{\Pi}^{1/2}$ and is written $\mathcal{J}_j = \mathcal{J}(A^{(j)})$. All current norms below are ordinary Hilbert norms. Applying an additional Q_{Π} norm would count the production matrix twice.

Put

$$Y_j = P_j \mathcal{J}_j, \quad b_j = d_j \mathcal{J}_j, \quad c_j = P_{j-1}(\mathcal{J}_j - \mathcal{J}_{j-1}).$$

The FAR separation is C , chosen at least as large as the harmless support constant that appears below.

3 The exact stopped-current representation

Theorem 3.1 (changing-current telescope)

For every $j_0 \leq j \leq J$,

$$\boxed{Y_j - Y_{j-1} = b_j + c_j.} \quad (3.1)$$

Consequently, for $n \geq j_0$,

$$\boxed{B_n := \sum_{j=j_0}^n b_j = Y_n - Y_{j_0} - \sum_{j=j_0}^n c_j.} \quad (3.2)$$

Proof. Insert and subtract $P_{j-1}\mathcal{J}_j$:

$$\begin{aligned} Y_j - Y_{j-1} &= (P_j - P_{j-1})\mathcal{J}_j + P_{j-1}(\mathcal{J}_j - \mathcal{J}_{j-1}) \\ &= b_j + c_j. \end{aligned}$$

Summing gives (3.2). No limiting operation is involved. $\square$

Because b_j is a martingale difference, the vectors $\Pi_m b_j$ are pairwise orthogonal in $L^2(\Omega; H)$ for fixed m . Define the deterministic index

$$n_C(m) = \min\{J, m - C\},$$

and interpret every sum ending below j_0 as zero.

Theorem 3.2 (FAR wedge as a stopped square)

The R-080 FAR quantity satisfies the exact identity

$$\begin{aligned} S_C &:= \sum_{j=j_0}^J \sum_{m \geq j+C} \mathbb{E} \|\Pi_m b_j\|^2 \\ &= \sum_m \sum_{j=j_0}^{n_C(m)} \mathbb{E} \|\Pi_m b_j\|^2 \\ &= \sum_m \mathbb{E} \left\| \Pi_m \left[Y_{n_C(m)} - Y_{j_0} - \sum_{j=j_0}^{n_C(m)} c_j \right] \right\|^2. \end{aligned} \quad (3.3)$$

Proof. Exchange the two deterministic sums. Martingale orthogonality turns the inner sum of squares into the square of $B_{n_C(m)}$. Apply (3.2). Since $n_C(m)$ is deterministic, no stopping-time hypothesis or optional-stopping theorem is used. $\square$

The predictable drift is load-bearing

For $Z_j^- = U_{j-1} + A^{(j-1)}$, $a_j = A^{(j)} - A^{(j-1)}$, and the R-079 heat-projected coefficient

$$\overline{M}_{j-1} = P_{\Sigma_{j:J}} M,$$

the drift has the exact two-channel form

$$\begin{aligned} c_j &= Q_{\Pi}^{1/2} [\overline{M}_{j-1}(Z_j^- + a_j) - \overline{M}_{j-1}(Z_j^-)]^T D Z_j^- \\ &\quad + Q_{\Pi}^{1/2} \overline{M}_{j-1}(Z_j^- + a_j)^T D a_j. \end{aligned} \quad (3.4)$$

The first line is predictable control-value injection and the second is predictable control-derivative injection. In the deterministic changing current example $\mathcal{J}_j = x_j$, every b_j is zero while $Y_n - Y_\ell = \sum_j c_j = x_n - x_\ell$. Thus deletion of c_j makes (3.2) false even before any analytic estimate is attempted.

4 The heat compensator and the three root channels

Let

$$Z = U_{j-1} + A^{(j)}, \quad g_j = U_j - U_{j-1},$$

and set

$$\overline{M}_j = P_{\Sigma_{j+1:J}} M, \quad \overline{M}_{j-1} = P_{\Delta \Sigma_j} \overline{M}_j.$$

R-079 expands a root exactly as

$$b_j = \iota_j + \nu_j + \kappa_j, \tag{4.1}$$

where

$$\begin{aligned} \iota_j &= Q_{\text{II}}^{1/2} \overline{M}_j (Z + g_j)^T D g_j, \\ \nu_j &= Q_{\text{II}}^{1/2} [\overline{M}_j (Z + g_j) - \overline{M}_j (Z)]^T D Z, \\ \kappa_j &= Q_{\text{II}}^{1/2} [\overline{M}_j - \overline{M}_{j-1}] (Z)^T D Z. \end{aligned} \tag{4.2}$$

These are derivative injection, raw value innovation, and heat compensator. For a common real-even covariance the root value $g_j(x)$ and root derivative $Dg_j(x)$ are independent. Therefore

$$\mathbb{E}_{j-1} \iota_j = 0, \quad \mathbb{E}_{j-1} \nu_j = -\kappa_j. \tag{4.3}$$

The centered object is $\nu_j + \kappa_j$, not ν_j alone.

The scalar fixture $M(x) = x^2$, with a centered root of variance σ , detects the error immediately:

$$\mathbb{E}_{j-1} \nu_j = \sigma DZ, \quad \kappa_j = -\sigma DZ. \tag{4.4}$$

This corrects any informal use of “centered value innovation” in which the heat compensator was not explicitly joined to it.

For the production Cartan quotient, write

$$F_A(z) = q_A(z)z, \quad \Phi_{A,j} = P_{\Sigma_{j+1:J}} F_A.$$

Up to the fixed factor -2 and the finite Q_{II} coordinates, the Cartan root is

$$\begin{aligned} b_{A,j}^{\text{Car}} &= -2 \{ \Phi_{A,j} (Z + g_j)^T D g_j \\ &\quad + [\Phi_{A,j} (Z + g_j) - \Phi_{A,j} (Z)]^T D Z \\ &\quad + [\Phi_{A,j} - \Phi_{A,j-1}] (Z)^T D Z \}. \end{aligned} \tag{4.5}$$

The safe centered split is

$$p_{A,j} = \Phi_{A,j} (Z)^T D g_j \tag{4.6}$$

and

$$\begin{aligned} r_{A,j} &= [\Phi_{A,j} (Z + g_j) - \Phi_{A,j} (Z)]^T (DZ + Dg_j) \\ &\quad + [\Phi_{A,j} - \Phi_{A,j-1}] (Z)^T DZ. \end{aligned} \tag{4.7}$$

Both $p_{A,j}$ and $r_{A,j}$ have conditional mean zero. The second object keeps the raw value innovation, nonlinear derivative injection, and heat compensator in one balanced block. It must not be factorised through $D_j A$; the R-081 witness already rules out that factorisation.

5 The uncontrolled FAR channel closes

Set $A^{(j)} \equiv 0$ and write $\mathcal{J}^0$ for the resulting current. It does not change with j , hence $c_j = 0$ and

$$B_n^0 = (P_n - P_\ell)\mathcal{J}^0. \quad (5.1)$$

The FAR loss is therefore

$$S_C^0 = \sum_m \mathbb{E} \left\| \Pi_m(P_{n_C(m)} - P_\ell)\mathcal{J}^0 \right\|^2. \quad (5.2)$$

The global R-050 conclusion $\mathcal{J}^0 \in \mathcal{C}^{-1/2-\kappa}$ does not by itself sum (5.2). The needed improvement is localised to the region $m \geq n + C$.

Lemma 5.1 (support-refined R-050 tail)

Fix

$$\frac{1}{3} < \alpha < \frac{1}{2}, \quad 0 < \beta < 3\alpha - 1.$$

For the heat-projected uncontrolled current and $m \geq n + C_0$,

$$\boxed{\mathbb{E} \|\Pi_m P_n \mathcal{J}^0\|_{L_x^2}^2 \leq C_{\alpha, \beta, e}, 2^{-2\beta m}.} \quad (5.3)$$

The constant is uniform in n and the ultraviolet cutoff.

Proof. Conditional expectation over future roots gives a smooth heat-averaged coefficient evaluated on the band-limited field U_n and paired with DU_n . The common even covariance eliminates the local coefficient–future derivative contraction. R-050 gives, uniformly in the target heat,

$$F_n(U_n) = T_{DF_n(U_n)}U_n + R_n, \quad R_n \in \mathcal{C}^{2\alpha}, \quad (5.4)$$

with all required second moments bounded independently of the cutoff.

The high factor in $T_{DF_n(U_n)}U_n$ is a block of U_n , so this term has spatial support at frequencies at most a fixed multiple of 2^n . After multiplication by DU_n , every low–high or resonant term whose high factor is U_n or DU_n has the same support. It vanishes when $m \geq n + C_0$.

In that relative-far region, the only possible high factor is the parilinearisation remainder R_n . Its low–high product with DU_n has regularity

$$(\alpha - 1) + 2\alpha = 3\alpha - 1 > 0. \quad (5.5)$$

The other Bony pieces again have their high factor in DU_n and vanish by support. Thus the relative-far part is bounded in $L^2(\Omega; \mathcal{C}^{3\alpha-1-})$. Littlewood–Paley projection gives (5.3) for every strict $\beta < 3\alpha - 1$. $\square$

The strict inequality is load-bearing. At the convenient value $\alpha = 2/5$, every $\beta < 1/5$ is admitted, but $\beta = 1/5$ is not claimed. Although α may approach $1/2$, no endpoint half derivative is claimed.

Theorem 5.2 (cutoff-uniform uncontrolled FAR)

For $C \geq C_0 + 1$ and every exponent in Lemma 5.1,

$$\boxed{S_C^0 \leq C_{\alpha, \beta, e, j_0}, 2^{-2\beta C}.} \quad (5.6)$$

Proof. For every nonempty stopped term, $n_C(m) = m - C$ unless the terminal cutoff has already been reached; in either case $m \geq n_C(m) + C_0$. Apply (5.3) to the moving endpoint and to the fixed endpoint $P_\ell \mathcal{J}^0$, use $\|x - y\|^2 \leq 2\|x\|^2 + 2\|y\|^2$, and sum the geometric series beginning at $m \geq j_0 + C$. The terminally saturated range is bounded by the same estimate and does not introduce a factor depending on J . $\square$

This theorem closes only the uncontrolled contribution. It does not turn a deterministic relative-gap estimate into the complete controlled root sum.

6 The exact controlled FAR target and a sufficient route

Define

$$\delta Y_j = P_j(\mathcal{J}_j - \mathcal{J}^0).$$

Subtracting the uncontrolled telescope from (3.2) gives

$$\sum_{j=j_0}^n d_j(\mathcal{J}_j - \mathcal{J}^0) = \delta Y_n - \delta Y_\ell - \sum_{j=j_0}^n c_j. \quad (6.1)$$

Thus the exact remaining controlled FAR theorem is the following estimate, not an estimate on any one channel separately:

$$\begin{aligned} \sum_m \mathbb{E} \left\| \Pi_m \left[\delta Y_{n_C(m)} - \delta Y_\ell - \sum_{j=j_0}^{n_C(m)} (c_j^{\text{val}} + c_j^{\text{der}}) \right] \right\|^2 \\ \leq K_e 2^{-\theta C} \left(1 + \mathbb{E}[X^{1/2} Y^{1/2}] \right), \quad \theta > 0. \end{aligned} \quad (\text{CFAR})$$

Here

$$X = \int \|h_s\|_2^2 ds, \quad Y = \|U + A^*\|_6^6.$$

The estimate must be uniform in the cutoff and in the declared regular control class. A large fixed C would then make the mixed coefficient small enough for the single Young allocation

$$K_e 2^{-\theta C} \mathbb{E} \sqrt{XY} \leq \eta \mathbb{E} X + \zeta \mathbb{E} Y. \quad (6.2)$$

The following exact lemma shows one possible analytic route.

Lemma 6.1 (orthogonal causal Carleson summation)

Suppose

$$f_j = \sum_{k \leq j} u_k,$$

and, for every fixed output shell m , the vectors $\Pi_m u_k$ are pairwise orthogonal in $L^2(\Omega; H)$. If

$$\mathbb{E} \|\Pi_m u_k\|^2 \leq 2^{-2s(m-k)} q_k, \quad m \geq k + C_0, \quad s > \frac{1}{2}, \quad (6.3)$$

then

$$\boxed{\sum_m \sum_{j \leq m-C} 2^j \mathbb{E} \|\Pi_m f_j\|^2 \leq C_s 2^{-2sC} \sum_k 2^k q_k}, \quad (6.4)$$

where one may take

$$C_s = \frac{1}{(1 - 2^{1-2s})(1 - 2^{-2s})}. \quad (6.5)$$

Proof. Orthogonality removes every $k \neq k'$ cross term. All terms are nonnegative, so the summations may be rearranged:

$$\begin{aligned} \sum_m \sum_{j \leq m-C} 2^j \|\Pi_m f_j\|_2^2 &\leq \sum_k q_k \sum_{j \geq k} 2^j \sum_{m \geq j+C} 2^{-2s(m-k)} \\ &= \frac{2^{-2sC}}{1 - 2^{-2s}} \sum_k 2^k q_k \sum_{r \geq 0} 2^{(1-2s)r}, \end{aligned}$$

which is (6.4). $\square$

The threshold is sharp for this triangular architecture. At $s = 1/2$, take only u_0 and prescribe $\mathbb{E}\|\Pi_m u_0\|^2 = 2^{-m}$. Then the contribution of each large m is bounded below by a positive constant because $\sum_{j \leq m-C} 2^j \asymp 2^{m-C}$. The truncated sum grows linearly.

If production analysis supplied

$$q_k \lesssim 2^{-4k} \mathbb{E}\|h_k\|_2^2,$$

then

$$\sum_k 2^k q_k \lesssim 2^{-3j_0} \mathbb{E}X, \quad (6.6)$$

so this channel would use the Cameron–Martin budget once. The missing step is precisely a causal, adapted, balanced input-scale decomposition with $s > 1/2$ for the combined stopped endpoint and predictable drift, retaining (4.7). R-081 shows that such a decomposition cannot be built from $D_j A$ alone. Lemma 6.1 is therefore a verified sufficient condition, not the complete controlled FAR theorem.

7 Global production current coordinates

Write the field and tangent as

$$\Psi = (u_1, u_2, \chi), \quad \dot{\Psi} = (v_1, v_2, w),$$

and put

$$r = |u|^2, \quad \rho = r + |\chi|^2, \quad d = \rho + e, \quad dr = 2 \operatorname{Re}(u^* v), \quad d\rho = 2 \operatorname{Re}(u^* v + \bar{\chi} w).$$

For the Pauli generators let

$$m_A = u^* \sigma_A u, \quad J_A = dm_A, \quad K_A = dm_A - m_A \frac{d\rho}{d}.$$

R-075 gives

$$Q_{\text{II}} = \begin{pmatrix} a & b \\ b & c \end{pmatrix}, \quad a = \frac{9}{500P}, \quad b = \frac{3}{400P}, \quad c = \frac{3}{320P}, \quad P = 4 + 10^{-12}. \quad (7.1)$$

Define the derived constants

$$c_0 = \frac{3}{250P}, \quad c_1 = \frac{243}{8000P}, \quad \alpha = \frac{5}{9}. \quad (7.2)$$

Then

$$T = \begin{pmatrix} \sqrt{c_0} & 0 \\ \sqrt{c_1}(1 - \alpha) & \sqrt{c_1}\alpha \end{pmatrix}, \quad T^T T = Q_{\text{II}}. \quad (7.3)$$

Thus $T(J_A, K_A)^T = (\sqrt{c_0}J_A, \sqrt{c_1}L_A)^T$, where

$$L_A = (1 - \alpha)J_A + \alpha K_A = J_A - \alpha m_A \frac{d\rho}{d}. \quad (7.4)$$

The resulting six-coordinate current is a fixed orthogonal representation of the R-079 $Q_{\text{II}}^{1/2}$ current and preserves its individual conditional root blocks.

Theorem 7.1 (global Pauli–Fierz square)

Let

$$h = u_1 v_2 - u_2 v_1$$

and define

$$\Xi(\Psi, \dot{\Psi}) = \begin{pmatrix} \sqrt{c_0} dr \\ \sqrt{c_1} \left(dr - \alpha \frac{r}{d} d\rho \right) \\ 2\sqrt{c_0 + c_1} \operatorname{Re} h \\ 2\sqrt{c_0 + c_1} \operatorname{Im} h \end{pmatrix}. \quad (7.5)$$

Then, globally including $r = 0$,

$$\sum_{A=1}^3 |(J_A, K_A)|_{Q_{\text{II}}}^2 = |\Xi(\Psi, \dot{\Psi})|^2. \quad (7.6)$$

The physical energy has the outer factor $1/2$.

Proof. The Pauli completeness relation gives the chart-free Fierz identities

$$\sum_A J_A^2 = (dr)^2 + 4|h|^2, \quad (7.7)$$

and

$$\sum_A L_A^2 = \left(dr - \alpha \frac{r}{d} d\rho \right)^2 + 4|h|^2. \quad (7.8)$$

Equation (7.3) diagonalises each A block. Summing $c_0 J_A^2 + c_1 L_A^2$ and applying (7.7)–(7.8) gives (7.6). The determinant h avoids division by r , so the formula remains valid at $u = 0$. $\square$

In the six-real ordering

$$(\operatorname{Re} \psi_1, \operatorname{Re} \psi_2, \operatorname{Re} \psi_3, \operatorname{Im} \psi_1, \operatorname{Im} \psi_2, \operatorname{Im} \psi_3),$$

write $C(z)y = \Xi(z, y)$. Polarisation of (7.6) gives

$$B(z) = C(z)^T C(z), \quad (7.9)$$

where C is 4×6 . When a complex covariance is converted to these real coordinates, the convention is $\Gamma_{\mathbb{R}} = \frac{1}{2} \operatorname{realify}(D_{\mathbb{C}})$.

Only the second row of (7.5) is rational. If P_{dbl} is the projector onto the doublet coordinates, its coefficient is

$$g(z) = P_{\text{dbl}} z - \alpha \frac{r}{d} z.$$

The elementary estimates $r/d \leq 1$ and $|z| |\nabla(r/d)| \leq 1$ yield

$$\|Dg\|_{\text{op}} \leq 1 + 2\alpha = \frac{19}{9}. \quad (7.10)$$

Thus the complete current reduces to three linear rows and one globally Lipschitz rational row. This is a genuine simplification, but it is not a signed NEAR estimate.

Why the four-coordinate compression is not rootwise

The map from the fixed six-current to (7.5) depends on the state. It preserves pointwise norm, B , and the Wick trace, but conditional expectation does not commute with that state-dependent compression. The independent two-root regression has equal pointwise and terminal energies but different first- and second-root Doob squares. Therefore R-079 rootwise bounds may be rerun directly in Ξ space, but may not be transported block by block from the fixed six-current merely from (7.6).

For a terminal pair Ξ_* and baseline Ξ_0 , let

$$R_j = d_j(\Xi_* - \Xi_0).$$

The exact resummed current identity is

$$\begin{aligned} \mathbb{E}[\mathcal{W}_J(A^*) - \mathcal{W}_J(A^0)] &= \frac{1}{2}\mathbb{E}(\|P_\ell \Xi_*\|^2 - \|P_\ell \Xi_0\|^2) - \frac{1}{2}\mathbb{E}(B_* - B_0) : \Gamma_\ell \\ &\quad + \sum_j \left[\mathbb{E}\langle d_j \Xi_0, R_j \rangle + \frac{1}{2}\mathbb{E}\|R_j\|^2 - \frac{1}{2}\mathbb{E}(B_* - B_0) : \Delta \Gamma_j \right]. \end{aligned} \quad (7.11)$$

This equals the re-summed R-079 injected/future-current algebra. The positive Doob square does not control the mixed baseline, Wick trace, paid subtraction, or R-063 forest.

8 Heat lifting and the exact covariance criterion

Target heat must act on the Gram matrix, not on C before squaring. The correct enlarged Hilbert coordinate is

$$\Xi_\Sigma(z, y)(r) = C(z + r)y \in L^2(\mu_\Sigma; \mathbb{R}^4), \quad (8.1)$$

so that

$$\|\Xi_\Sigma(z, y)\|^2 = y^T (P_\Sigma B)(z) y. \quad (8.2)$$

In contrast,

$$P_\Sigma B - (P_\Sigma C)^T (P_\Sigma C) = \mathbb{E}[(C - P_\Sigma C)^T (C - P_\Sigma C)] \succeq 0. \quad (8.3)$$

At $z = 0$ under the symmetric law $r = \pm te_1$, one has $P_\Sigma C = 0$ but $P_\Sigma B = B(te_1) \neq 0$. This also shows why changing heat covariance across roots requires the compensator in Section 4.

Theorem 8.1 (conditional square–trace identity)

Let $C(z)$ and a derivative control c be measurable with respect to a sigma-algebra $\mathcal{A}$. For a fresh derivative vector G , put

$$\mu = \mathbb{E}[G \mid \mathcal{A}], \quad \Sigma_c = \text{Cov}(G \mid \mathcal{A}), \quad B = C^T C.$$

Then

$$\begin{aligned} \mathbb{E} \left[\frac{1}{2} |C(z)(G + c)|^2 - \frac{1}{2} \text{Tr}(B(z)\Gamma) \mid \mathcal{A} \right] \\ = \boxed{\frac{1}{2} |C(z)(c + \mu)|^2 + \frac{1}{2} \text{Tr}[B(z)(\Sigma_c - \Gamma)]}. \end{aligned} \quad (8.4)$$

Proof. Write $G + c = (G - \mu) + (c + \mu)$. The conditional mixed term vanishes. The conditional expectation of the centered square is $\text{Tr}(B\Sigma_c)$, which gives (8.4). $\square$

If $\mu = 0$ and $\Sigma_c = \Gamma$, (8.4) is the nonnegative square $\frac{1}{2}|C(z)c|^2$. This closes the predictable coefficient/fresh derivative subbranch. General later feedback changes μ or Σ_c ; the remaining trace defect is signed. The two conditional laws $G = 0$ and $G = \pm\sqrt{2}$ relative to $\Gamma = 1$ give defects $-1/2$ and $+1/2$. Accordingly, covariance preservation is a hypothesis to prove, not a consequence of adaptedness.

9 Moving shell edges are a signed repartition

Let $L_j \leq U_j$ be nested exact orthogonal spatial projectors commuting with the target quadratic form. At a fixed complete current c , divide space into low L_j , near $U_j - L_j$, and high $I - U_j$. In the forward partition convention the edge fluxes are

$$\begin{aligned}\Phi_{\text{low}} &= +\frac{1}{2}\|\Delta L_j c\|_Q^2, \\ \Phi_{\text{near}} &= +\frac{1}{2}\|\Delta U_j c\|_Q^2 - \frac{1}{2}\|\Delta L_j c\|_Q^2, \\ \Phi_{\text{high}} &= -\frac{1}{2}\|\Delta U_j c\|_Q^2.\end{aligned}\tag{9.1}$$

Therefore

$$\Phi_{\text{low}} + \Phi_{\text{near}} + \Phi_{\text{high}} = 0.\tag{9.2}$$

All signs reverse in the backward/loss convention. This identity is the sharp-shell repartition of the R-069 pure-control telescope. It does not create a positive absolute pair-high term and does not control smooth Littlewood–Paley commutators, the covariance trace, paid terms, or the forest.

10 Evidence map: successes, failed routes, and reuse

The proof search is recorded by mathematical route rather than only by its successful endpoint.

1. *Direct rootwise FAR summation.* R-081 found the critical repeated $X^{1/2}Y^{1/2}$ loss. Verdict: failed as a one-use argument. Reuse: it identified the need to sum orthogonally before applying Young.
2. *Stopped-current reordering.* The deterministic wedge and martingale orthogonality give (3.3). Verdict: exact success. Boundary: the moving endpoint and c_j remain inside the norm.
3. *Dropping predictable drift.* A deterministic changing current has $b_j = 0$ and nonzero c_j . Verdict: falsified. Reuse: (CFAR) now keeps both drift channels explicitly.
4. *Calling the raw value innovation centered.* The $M(x) = x^2$ fixture gives (4.4). Verdict: falsified. Reuse: value innovation and heat compensator are joined in the balanced block (4.7).
5. *Using only the global R-050 negative regularity.* That estimate does not sum the stopped FAR wedge. Verdict: insufficient. Reuse: spatial support isolates the $3\alpha - 1$ parilinear remainder and proves Lemma 5.1.
6. *Uncontrolled support-refined FAR.* Lemma 5.1 and the stopped identity give Theorem 5.2. Verdict: success, uniformly in the cutoff.
7. *A factorised controlled coefficient with causal input scales.* Orthogonal Carleson summation succeeds for $s > 1/2$ and fails sharply at $s = 1/2$. Verdict: conditional success. Missing input: the production balanced decomposition for the full endpoint–drift combination.
8. *Factoring the nonlinear current through $D_j A$.* The R-081 upper-triangular Jensen witness has zero $D_j A$ and nonzero coefficient innovation. Verdict: falsified. Reuse: retain the secant, derivative, value, and heat terms jointly.
9. *Diagonalising the production current.* The fixed bridge (7.3) and Pauli completeness give the global four-coordinate square (7.5). Verdict: exact success, including $r = 0$.

10. *Transporting individual root blocks through Pauli–Fierz compression.* The state-dependent map redistributes two-root Doob energy. Verdict: falsified. Reuse: apply Doob directly in Ξ space and preserve the complete packet.
11. *Squaring the heat-averaged coordinate.* Formula (8.3) and the symmetric heat witness show that $(P_\Sigma C)^T(P_\Sigma C)$ loses a positive variance. Verdict: falsified. Reuse: lift $C(z+r)$ into the heat Hilbert space.
12. *Conditional covariance preservation.* The fresh independent subbranch becomes a positive square by (8.4); future feedback has a signed covariance defect. Verdict: exact conditional success, general closure absent.
13. *Using moving edge flux as a new payment.* The three regions cancel exactly by (9.2). Verdict: book-keeping success, positivity claim falsified. Reuse: preserve the signed complete current telescope.

No new negative-result card is needed for the recurring nonlinear, covariance, or absolute pair-high obstructions: the existing A13 negative records already govern them. The new centering correction and compression regression are retained in both executables and this evidence map so that they cannot silently regress.

11 Remaining route to Sector A

After R-082 the narrow proof order is:

$$\boxed{\text{controlled (CFAR)}} \longrightarrow \boxed{\text{complete signed NEAR square–trace–forest}} \longrightarrow \boxed{\text{overlap-stable progression}} \quad (11.1)$$

followed by

$$\boxed{\text{controlled-shell one-use}} \longrightarrow \boxed{q = 10/9 \text{ Nelson synthesis}} \longrightarrow \boxed{\text{Sector-A closure audit}}. \quad (11.2)$$

The next controlled-FAR attempt should prove the balanced causal input-scale decomposition required after Lemma 6.1, with the stopped endpoint and both predictable drifts treated together. In parallel, the NEAR proof should use the four-row coordinate (7.5) but retain the heat Hilbert lift, the mixed baseline, the covariance trace, the R-063 forest, and the existing paid subtraction. The only rational row has the global Lipschitz ceiling 19/9; the other three rows are linear. Any positivity step must state its measurability and conditional-covariance hypotheses.

The arithmetic target remains conditional:

$$\varepsilon_V = \frac{9}{20}, \quad q = \frac{1}{2\varepsilon_V} = \frac{10}{9}, \quad q - \frac{11}{10} = \frac{1}{90}. \quad (11.3)$$

These equalities do not establish one-use, Nelson, an interacting measure, a removal limit, or Sector-A closure.

12 Devil’s-advocate self-test

1. **Objection:** (3.3) double-counts Q_{II} or uses optional stopping. **DISMISSED.** The current already carries $Q_{\text{II}}^{1/2}$, the norm is the ordinary Hilbert norm, and $n_C(m)$ is a deterministic index.
2. **Objection:** the endpoint formula may omit the changing-control drift. **VALID WITH MITIGATION.** The omission is false; (3.4), the deterministic counterfixture, (CFAR), and both executables retain it.

3. **Objection:** Lemma 5.1 follows from the global R-050 regularity alone. **UPHELD against that argument.** The proof instead uses the band support of every rough high factor and the positive $3\alpha - 1$ remainder. Only strict $\beta < 3\alpha - 1$ is asserted.
4. **Objection:** the Carleson lemma proves controlled FAR. **UPHELD.** It is only a sufficient summation lemma. The causal production decomposition (6.3), jointly for endpoint and drift, is open.
5. **Objection:** the four-coordinate norm identity transports all R-079 root blocks. **UPHELD.** Only the fixed six-current does so. The Pauli–Fierz compression is state-dependent and the independent two-root fixture explicitly redistributes root energy.
6. **Objection:** heat averaging commutes with the square root coordinate. **UPHELD.** Formula (8.3) gives the missing positive variance. The heat-lifted coordinate (8.1) is used instead.
7. **Objection:** complete-current positivity implies translated convexity. **UPHELD.** The existing production Bregman fixture has negative second derivative. The square, trace, mixed, paid, and forest terms must remain together.
8. **Objection:** (9.1) supplies a new positive pair-high payment. **UPHELD.** The three edge fluxes sum to zero. It is only a signed repartition of the pure-control telescope.

External adversarial review is invited specifically for the support-refined step in Lemma 5.1, the orthogonality hypothesis in Lemma 6.1, the real/complex covariance convention under (7.9), and the conditional covariance hypothesis in (8.4).

13 Reproduction and independent audit

The primary executable checks the changing-current telescope, martingale orthogonality, stopped FAR square, nonzero predictable drift, quadratic heat compensator, support and exponent ledgers, the orthogonal Carleson threshold, the global Fierz coordinate, complete-current Doob variance, conditional square–trace identity, Bregman negative control, moving edge flux, and honesty flags. The non-importing executable uses an alternate finite filtration, random Fourier data, direct NumPy Gram matrices, the two-root state-dependent compression regression, the symmetric heat witness, and independent covariance and edge fixtures.

Run from the repository root:

```
python codes/foundations/
a13_classii_stopped_current_far_complete_current_near_
reduction_verify.py
```

The child contracts are primary 54/54 and non-importing independent 52/52. The integrated verifier pins R-050, R-063, R-075, R-079, R-080, and R-081; checks the note, PDF, manifest, run, result, roadmap, gate, task, claim, exploration, and honesty surfaces; and fails closed if controlled FAR, complete signed NEAR, progression, one-use, Nelson, measure, or Sector-A closure is promoted.

Result footer

Result ID: A13-CLASSII-STOPPED-CURRENT-FAR-COMPLETE-CURRENT-NEAR-COORDINATE-REDUCTION

Precise statement: The FAR wedge is exactly one deterministic stopped-current square retaining the moving endpoint and predictable value/derivative drift. The raw value innovation centers only with its heat compensator. The uncontrolled production FAR loss decays

cutoff-uniformly as $2^{(-2 \beta C)}$ for every $\beta < 3 \alpha - 1$. An orthogonal causal Carleson lemma has the sharp $s > 1/2$ threshold. The complete production current has a global four-coordinate Pauli--Fierz square and an exact conditional-covariance positivity criterion, but the compression is not rootwise and future covariance defects remain.

Scope: Finite cutoff; fixed positive floor; common even regulator; exact products; pinned production frame; constants cutoff-uniform only where stated.

Dependencies: R-050; R-063; R-069; R-070; R-075; R-079; R-080; R-081.

Evidence grade: ANALYTIC, EXACT, EXECUTED.

Reproduction command: `python codes/foundations/a13_classii_stopped_current_far_complete_current_near_reduction_verify.py`

Expected output: Primary 54/54; independent 52/52; manifest-pinned integrated and aggregate PASS; exit zero.

Falsification gate: Failure of the stopped identity, drift or heat centering, support-refined tail, Carleson threshold, Fierz/Gram identity, compression regression, heat lift, covariance identity, edge cancellation, evidence hash, PDF contract, or honesty flag.

Tier before / after: T4 / T4; no promotion.

No-overclaim statement: No controlled FAR theorem, complete signed NEAR theorem, overlap-stable progressive lower bound, controlled-shell one-use, Nelson bound, measure theorem, removal limit, or Sector-A closure is proved.

Next required action: Prove balanced causal controlled CFAR with $s > 1/2$, then the complete heat-lifted signed NEAR packet and overlap-stable progression before one-use synthesis.